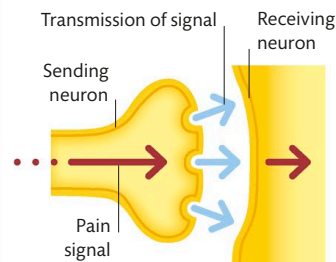


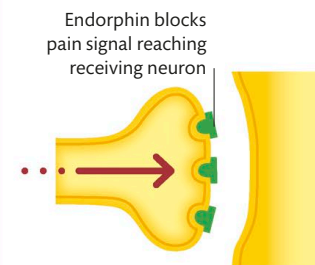


NATURAL PAIN RELIEF

The body releases its own chemicals, called endorphins and enkephalins, to dampen the pain signals. They bind to receptors on the nerve endings, preventing further transmission of pain signals.



PAIN SIGNAL TRANSMITTED



BLOCKED PAIN SIGNAL

Frontal cortex plays role in anticipating and controlling pain

Somatosensory cortex identifies intensity, location, and type of pain

Limbic system is responsible for emotional and behavioral reaction to pain

Reticular formation modulates pain signals

Thalamus relays signals to different areas of brain

4 Pain signals processed

The signal continues to the thalamus, which distributes impulses to the cortex and other areas responsible for emotion, attention, and assessing the significance of the pain.

Nerve fibers descending from brain intercept and modify ascending pain signals

5 Alleviating pain

Descending signals travel back down from the brain to intercept the pain signals (see box, right). These trigger the release of natural painkillers by the brain stem and spinal cord that reduce pain signals.

Pain signals travel up spinal cord

Spinal cord signal bypasses brain

3 Pain signals reach the spinal column

The nerve bundle enters the spinal cord through the dorsal horn. Pain signals pass across to the other side of the spinal cord for their onward journey to the brain.

SPINAL CORD

DORSAL HORN

Most nerve bundles enter at back of spine, known as dorsal horn